Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/28/2016
PI Name:	Weitz, David Ph.D.		
Project Title:	Physics of Colloids in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02138	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Fluid Physics Rapid Turn Around Flight Opportunity
Start Date:	11/01/2013	**End Date:**	09/30/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lu, Peter Ph.D. (Harvard University)		
Grant/Contract No.:	NNX13AQ48G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continues "Physics of Colloids in Space," grant # NNX08AE09G. We will study the general collective behavior of colloidal particles, including the structures they form and the dynamics of the associated processes. As a result of longterm support from NASA, we have developed significant expertise in the formulation and characterization of systems of hard-sphere colloidal particles in a variety of solvents, and a host of techniques in the laboratory to characterize their structure and behavior, including several forms of microscopy, notably laser-scanning confocal microscopy, optical macro-photography, and laser-light scattering. Using chemistry again developed over the course of many years and supported by NASA, we can very carefully control the interaction potential between these colloidal particles. We can make the colloidal particles hard-spheres, interacting only insofar as to repel when contacting each other. Through the mechanism of depletion, we can make the particles attractive, and can		

Task Description:	exert very precise control over the range and strength of this attraction. In aqueous solvents, we can carefully control the screening-length, thereby controlling the range of repulsion of charged particles; and we can do the same with micelles in non-aqueous solvents. We have also developed new sample chamber technology, to improve the optics and mechanics for confocal microscopy, as well as control environmental factors and even exert electric fields. All of these ground-based studies have provided core technologies that we expect to apply to microgravity experiments, in particular the Light Microscopy Module/Advanced Colloids Experiment (LMM/ACE) set of experiments that involve microscopy aboard the International Space Station (ISS). The main goal of our work on colloids is to continue to explore fundamental science through the behavior of these well characterized systems, and apply the lessons directly to very similar systems in the lab with important industrial applications. Furthermore, we are using the same technology and techniques to explore materials beyond rigid spherical colloids that diffuse: in particular, self-propelled “colloids” in the form of swimming bacteria; flexible and deformable “colloids” in the form of micron-scale emulsion droplets, and the dynamics of charged systems in the presence of large external fields. This provides a way to apply directly the expertise we have developed to design and support ISS-based research, and has the potential to give significant new insight and direction for future experiments in microgravity. The work that we will do for the next 5 years primarily involves supporting our current set of experiments currently onboard the ISS, Binary Colloidal Alloy Test (BCAT) BCAT-3, BCAT-4, BCAT-5, and BCAT-6 (subsequently referred to collectively as BCAT), as well continuing to develop the hardware, software, and samples to launch as part of the set of upcoming experiments ACE-M and ACE-C (subsequently referred to collectively as ACE). To do this properly, we must conduct a range of ground-based experiments, which allows the development of relevant technologies and expertise in order to successfully conduct flight experiments aboard the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These experiments are extremely important in determining the longer term shelflife stability of commercial products, and we plan to continue our work with some of our industrial collaborators, including Dr. Matthew Lynch at Procter and Gamble, to extend the utility of what we learn to real products. Our studies could also contribute significant understanding to the stability of many products that will likely be used for extended stays in space.
Task Progress:	We have made significant progress toward a number of goals associated with the colloids effort in microgravity in 2015, and have completed and/or exceeded all of our required deliverables: 1. We have made available the requisite number of test samples for the development of ACE confocal microscopy experiments. We formulated and delivered new samples to ZIN Technologies, who has confirmed delivery during the first week of August, 2015, as requested and on schedule. The volumes we delivered are enough to make many dozens of test engineering samples. These samples also add to the dozens of samples already in ZIN’s and NASA’s possession, which were originally designed and optimized for confocal microscopy, and using the ground-based National Science Foundation (NSF)-funded confocal microscopes we have in our laboratory. 2. We have published another colloid peer-reviewed research paper describing novel behavior of re-entrant crystallization in charged colloid systems, using materials and systems similar to what we have launched on orbit. This NASA-sponsored ground-based investigation may lead to an understanding of the general behavior of charged colloids in non-aqueous solvents, with a number of applications that may improve life on Earth, including the development of electronic paper displays (e.g., E-ink corporation has related technology), and the longer-term degradation of performance in industrial and motor oils. The paper notes properly our NASA funding. 3. We have successfully launched, operated, and concluded the ACE-M3 experiment, including space operations and analysis. 4. We have presented new scientific results from ACE-M2 and BCAT-KP on a poster at the 4th ISS R&D conference in Boston, MA, held on 7-9 Jul 2015. We are delighted by the overwhelmingly positive reception to our findings from NASA staff, management, and crew members. In particular, Bill Gerstenmaier, Associate Administrator for Human Exploration and Operations, called our results “spectacular.” Mike Suffredini, manager of the International Space Station program, said what we had done was “awesome.” Several former astronaut crew members, including Dan Tani, Carl Walz, and Cady Coleman, were excited by the extensions to the earlier BCAT experiments, which they have all worked on aboard ISS. We also presented the results extensively with Kirt Costello, deputy to Chief Scientist Julie Robinson, and spent an extended amount of time with Vic Cooley, LIS for several recent past increments, going over our scientific results, and giving him a tour of our lab at Harvard, to show ongoing ground-based work relating to ACE and BCAT. 5. To comply with the deliverable in our grant augmentation, we have evaluated the current status of the ACE confocal microscope system, and the recent proposed upgrades that are scheduled to launch in the upcoming year to the ISS. 6. To comply with the deliverable in our grant augmentation, we provide several documented software solutions for removing the overall motion between two images, and apply this to 2D time series and 3D image stacks. The software and documentation are provided under a GPL version 3 open-source license, so that it can be redistributed to other researchers and anyone around the world, furthering the goal of open science. That section of the report, deliberately, has no proprietary markings, to allow free redistribution for non-profit and scientific research use. The software is capable of using parallel architectures, including multi-core CPUs and GPUs, for enhanced performance on appropriately capable computer workstations, as specified in our deliverable requirement. 7. We have prepared the final report for the BCAT3-6 set of experiments, and removed all proprietary markings. We believe we have not only completed, but substantially exceeded, the requirements for our grant in 2015. We have made tremendous progress in publication of ground and flight results, delivering samples for future flight, and evaluating and assisting with development of those experiments. We look forward to continuing our successful partnership with NASA in upcoming years, with colloid experiments on the ground and in microgravity, aboard the International Space Station.
Bibliography Type:	Description: (Last Updated: 06/13/2025)

Articles in Peer-reviewed Journals	Kanai T, Boon N, Lu PJ, Sloutskin E, Schofield AB, Smallenburg F, van Roij R, Dijkstra M, Weitz DA. "Crystallization and reentrant melting of charged colloids in nonpolar solvents." Phys Rev E Stat Nonlin Soft Matter Phys. 2015 Mar;91(3):030301. Epub 2015 Mar 23. PubMed PMID: 25871032 ; http://dx.doi.org/10.1103/PhysRevE.91.030301 , Mar-2015